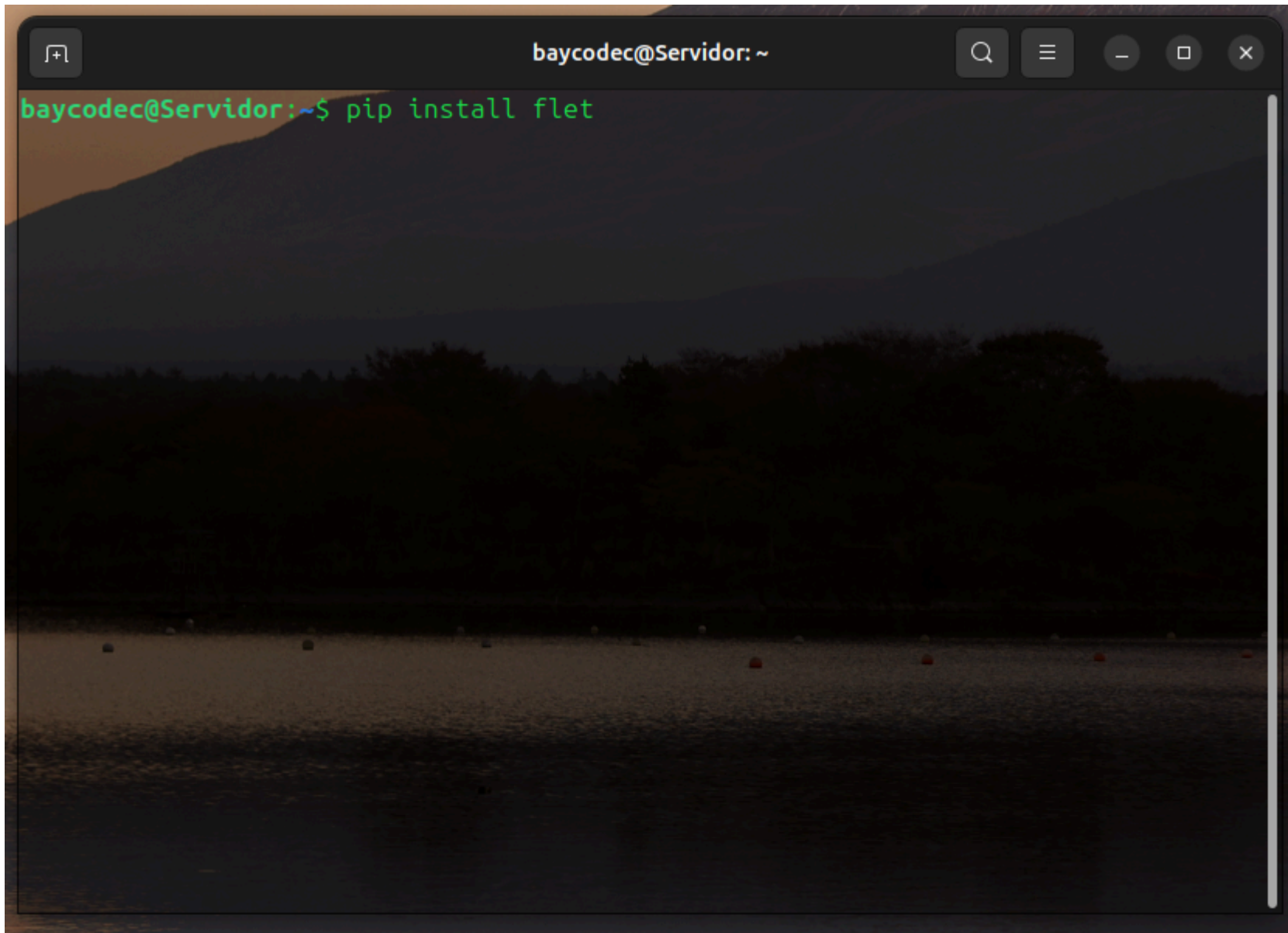


Método	Rendimiento	Curva de Aprendizaje	Reutilización de Código
Nativo (Kotlin)	Excelente	Media/Alta	Solo Android
Flutter / React Native	Muy Bueno	Media	Android e iOS
Híbrido (Ionic)	Regular	Baja (si sabes web)	Android, iOS y Web
Visual (App Inventor)	Limitado	Muy Baja	Limitado

Método	Android	iOS / iPod	Windows	Dificultad
Nativo	✓	✓	✓	Alta
Flutter / React Native	✓	✓	✓	Media
PWA / Web	✓	✓	✓	Baja
App Inventor / Blocks	✓	⚠ (Limitado)	✗	Muy Baja

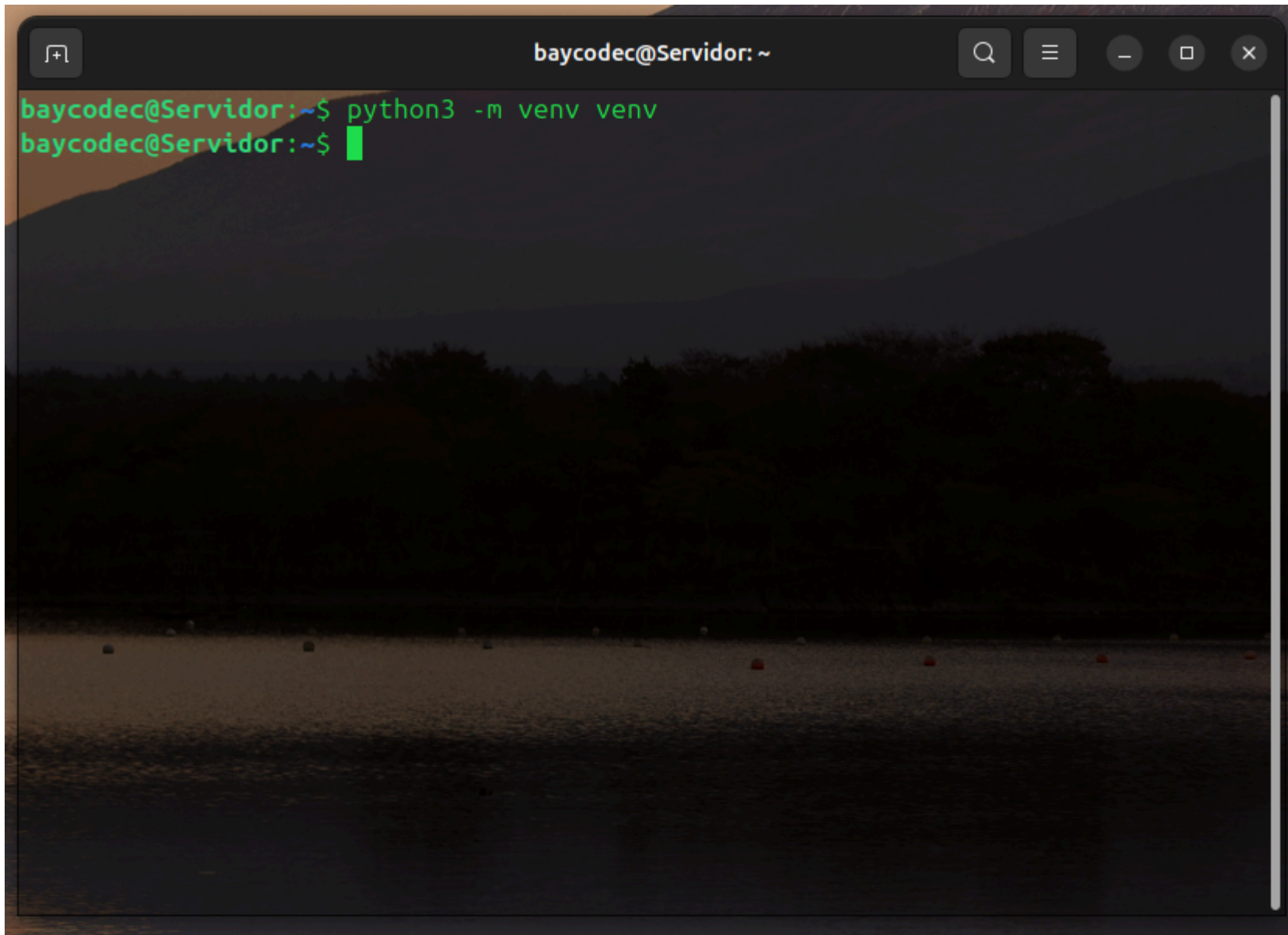
Categoría	Tecnología / Framework	Lenguaje Principal	Dispositivos Compatibles	Nivel de Rendimiento
Nativo	Android SDK	Kotlin / Java	Android (Celulares, Tablets)	Máximo
	Swift / Xcode	Swift	iOS (iPhone, iPad, iPod Touch)	Máximo
	WinUI 3 / WPF	C#	Windows (PC, Tablets Surface)	Máximo
Multipataforma	Flutter	Dart	Android, iOS, Windows, Web, Linux	Muy Alto
(Compilado)	React Native	JavaScript / TS	Android, iOS, Windows	Muy Alto
	.NET MAUI (Xamarin)	C#	Android, iOS, Windows, macOS	Alto
	Kotlin Multiplatform	Kotlin	Android, iOS (lógica compartida)	Máximo

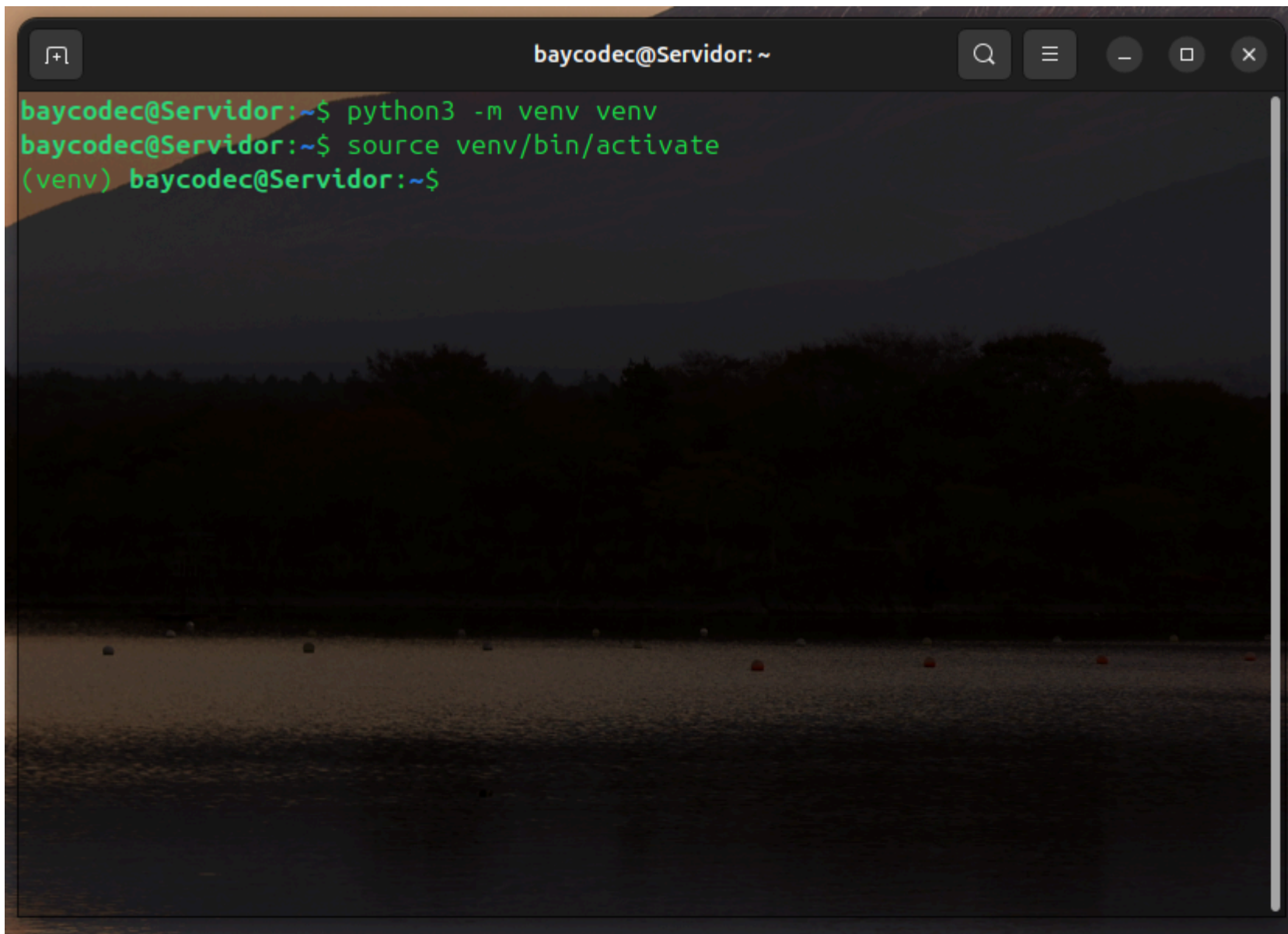
Basado en Python	Kivy	Python	Android, iOS, Windows, Linux	Medio
	BeeWare	Python	Android, iOS, Windows, macOS	Medio
Híbrido / Web	Ionic / Capacitor	HTML, CSS, JS	Android, iOS, Web	Medio
	PWA	Web Standard	Cualquier dispositivo con navegador	Depende de la red
Visual / Low-Code	App Inventor	Bloques (Lógica)	Android, iOS (Beta limitada)	Básico / Educativo
	Kodular / Thunkable	Bloques	Android, iOS	Básico / Profesional
	FlutterFlow	Visual / Dart	Android, iOS, Web	Alto



```
baycodec@Servidor: ~  
baycodec@Servidor:~$ pip install flet  
error: externally-managed-environment  
  
× This environment is externally managed  
└> To install Python packages system-wide, try apt install  
python3-xyz, where xyz is the package you are trying to  
install.  
  
If you wish to install a non-Debian-packaged Python package,  
create a virtual environment using python3 -m venv path/to/venv.  
Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make  
sure you have python3-full installed.  
  
If you wish to install a non-Debian packaged Python application,  
it may be easiest to use pipx install xyz, which will manage a  
virtual environment for you. Make sure you have pipx installed.  
  
See /usr/share/doc/python3.12/README.venv for more information.  
  
note: If you believe this is a mistake, please contact your Python installation  
or OS distribution provider. You can override this, at the risk of breaking your  
Python installation or OS, by passing --break-system-packages.  
hint: See PEP 668 for the detailed specification.  
baycodec@Servidor:~$
```

```
baycodec@Servidor: ~  
error: externally-managed-environment  
  
× This environment is externally managed  
└> To install Python packages system-wide, try apt install  
    python3-xyz, where xyz is the package you are trying to  
    install.  
  
    If you wish to install a non-Debian-packaged Python package,  
    create a virtual environment using python3 -m venv path/to/venv.  
    Then use path/to/venv/bin/python and path/to/venv/bin/pip. Make  
    sure you have python3-full installed.  
  
    If you wish to install a non-Debian packaged Python application,  
    it may be easiest to use pipx install xyz, which will manage a  
    virtual environment for you. Make sure you have pipx installed.  
  
    See /usr/share/doc/python3.12/README.venv for more information.  
  
note: If you believe this is a mistake, please contact your Python installation  
or OS distribution provider. You can override this, at the risk of breaking your  
Python installation or OS, by passing --break-system-packages.  
hint: See PEP 668 for the detailed specification.  
baycodec@Servidor:~$ python3 -m venv venv  
baycodec@Servidor:~$
```

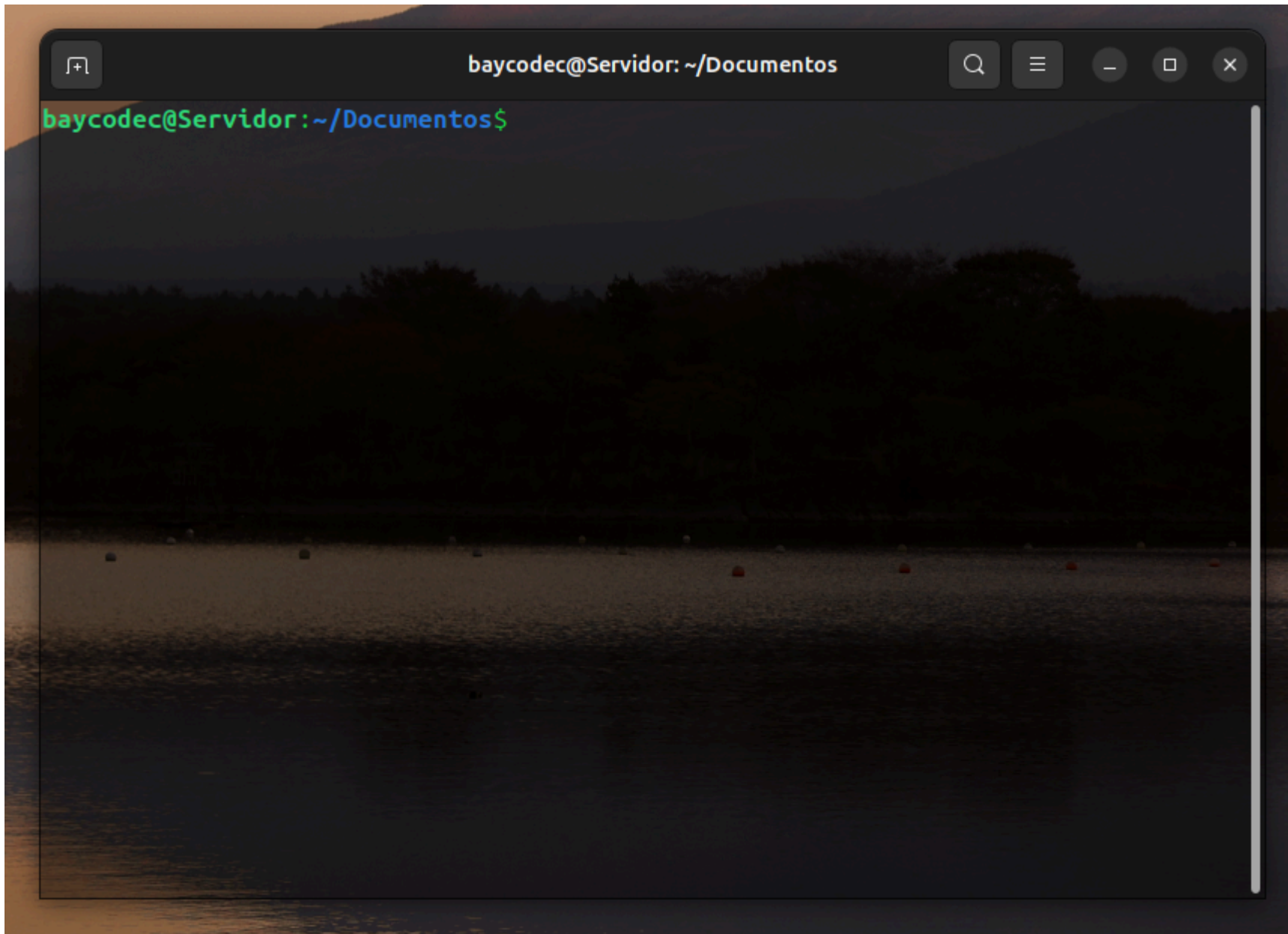


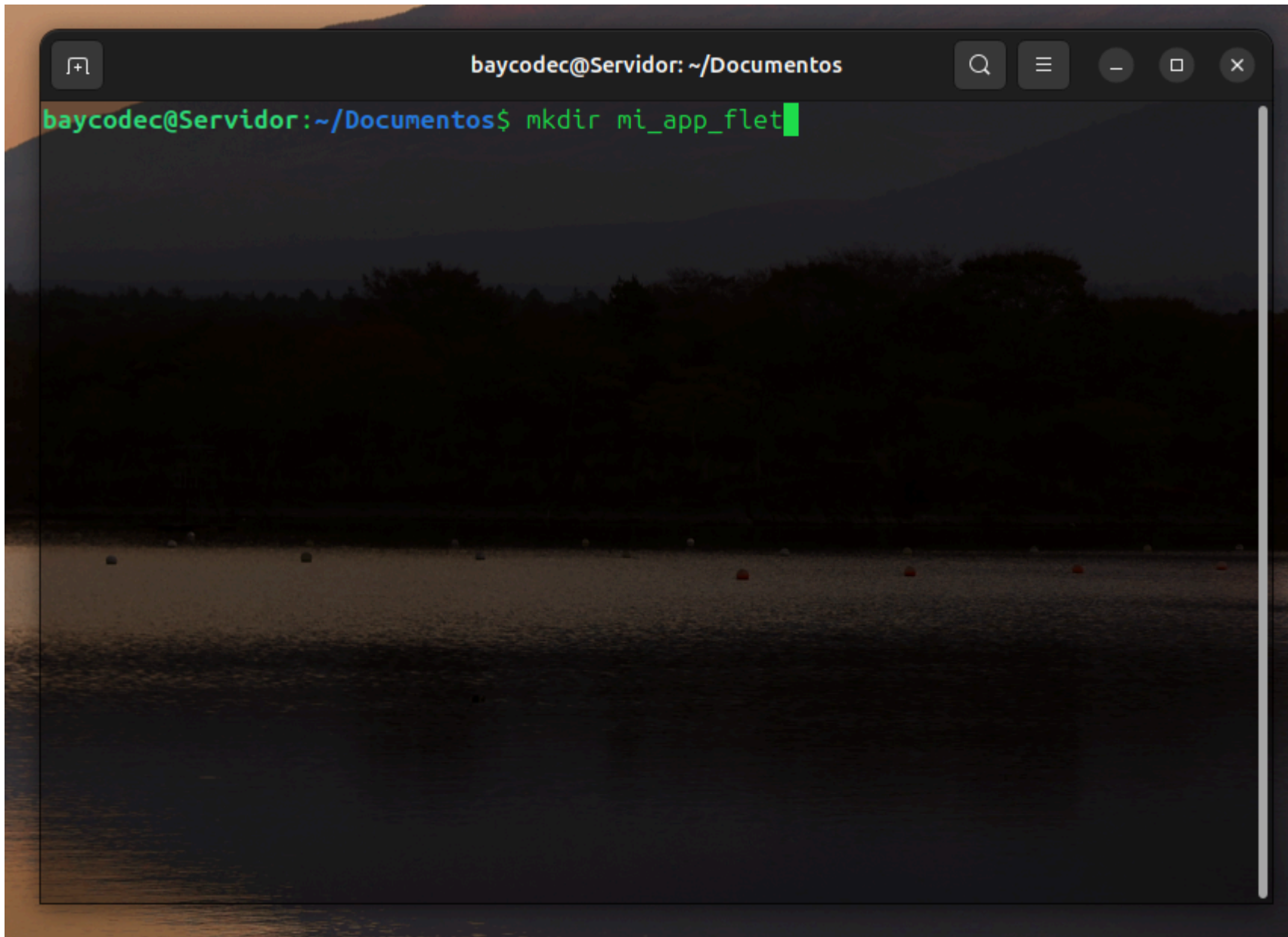


A terminal window titled "baycodec@Servidor: ~" with standard macOS window controls. The terminal shows the execution of two commands: "python3 -m venv venv" and "source venv/bin/activate". The output shows the prompt changing from "baycodec@Servidor:~\$" to "(venv) baycodec@Servidor:~\$". The background of the terminal is a dark, scenic image of a lake at night with mountains in the distance.

```
baycodec@Servidor:~$ python3 -m venv venv
baycodec@Servidor:~$ source venv/bin/activate
(venv) baycodec@Servidor:~$
```

```
baycodec@Servidor: ~  
579.8/579.8 kB 8.0 MB/s eta 0:00:00  
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)  
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)  
Downloading msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64  
.manylinux_2_28_x86_64.whl (427 kB)  
427.6/427.6 kB 23.4 MB/s eta 0:00:00  
Downloading oauthlib-3.3.1-py3-none-any.whl (160 kB)  
160.1/160.1 kB 41.0 MB/s eta 0:00:00  
Downloading repath-0.9.0-py3-none-any.whl (4.7 kB)  
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)  
Downloading anyio-4.13.0-py3-none-any.whl (114 kB)  
114.4/114.4 kB 30.6 MB/s eta 0:00:00  
Downloading idna-3.15-py3-none-any.whl (72 kB)  
72.3/72.3 kB 18.4 MB/s eta 0:00:00  
Downloading certifi-2026.5.20-py3-none-any.whl (134 kB)  
134.1/134.1 kB 52.0 MB/s eta 0:00:00  
Using cached h11-0.16.0-py3-none-any.whl (37 kB)  
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)  
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna,  
h11, certifi, repath, httpcore, anyio, httpx, flet  
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 htt  
pcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six  
-1.17.0 typing_extensions-4.15.0  
(venv) baycodec@Servidor:~$
```







baycodec@Servidor: ~/Documentos/mi_app_flet



```
baycodec@Servidor:~/Documentos$ mkdir mi_app_flet
```

```
baycodec@Servidor:~/Documentos$ cd mi_app_flet
```

```
baycodec@Servidor:~/Documentos/mi_app_flet$
```



baycodec@Servidor: ~/Documentos/mi_app_flet



```
baycodec@Servidor:~/Documentos$ mkdir mi_app_flet
baycodec@Servidor:~/Documentos$ cd mi_app_flet
baycodec@Servidor:~/Documentos/mi_app_flet$ python3 -m venv venv
baycodec@Servidor:~/Documentos/mi_app_flet$
```



baycodec@Servidor: ~/Documentos/mi_app_flet

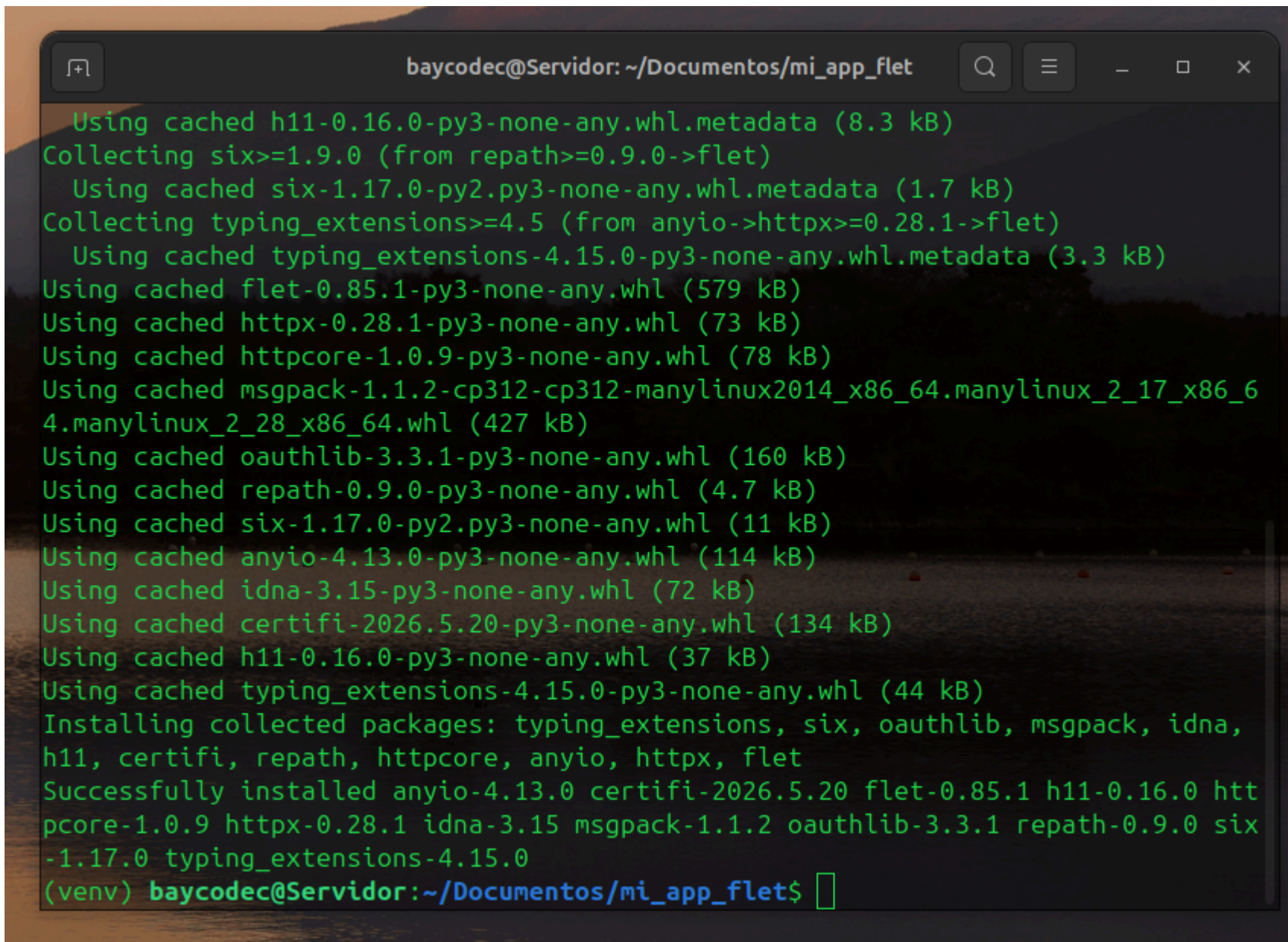


```
baycodec@Servidor:~/Documentos$ mkdir mi_app_flet
baycodec@Servidor:~/Documentos$ cd mi_app_flet
baycodec@Servidor:~/Documentos/mi_app_flet$ python3 -m venv venv
baycodec@Servidor:~/Documentos/mi_app_flet$ source venv/bin/activate
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$
```



```
baycodec@Servidor: ~/Documentos/mi_app_flet
baycodec@Servidor:~/Documentos$ mkdir mi_app_flet
baycodec@Servidor:~/Documentos$ cd mi_app_flet
baycodec@Servidor:~/Documentos/mi_app_flet$ python3 -m venv venv
baycodec@Servidor:~/Documentos/mi_app_flet$ source venv/bin/activate
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ pip install flet
Collecting flet
  Using cached flet-0.85.1-py3-none-any.whl.metadata (6.9 kB)
Collecting oauthlib>=3.2.2 (from flet)
  Using cached oauthlib-3.3.1-py3-none-any.whl.metadata (7.9 kB)
Collecting httpx>=0.28.1 (from flet)
  Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting repath>=0.9.0 (from flet)
  Using cached repath-0.9.0-py3-none-any.whl.metadata (899 bytes)
Collecting msgpack>=1.1.0 (from flet)
  Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (8.1 kB)
Collecting anyio (from httpx>=0.28.1->flet)
  Using cached anyio-4.13.0-py3-none-any.whl.metadata (4.5 kB)
Collecting certifi (from httpx>=0.28.1->flet)
  Using cached certifi-2026.5.20-py3-none-any.whl.metadata (2.5 kB)
Collecting httpcore==1.* (from httpx>=0.28.1->flet)
  Using cached httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Collecting idna (from httpx>=0.28.1->flet)
  Using cached idna-3.15-py3-none-any.whl.metadata (7.7 kB)
```

```
mkdir mi_app_flet
cd mi_app_flet
python3 -m venv venv
source venv/bin/activate
pip install flet
```

A terminal window titled 'baycodec@Servidor: ~/Documentos/mi_app_flet' with standard window controls. The terminal displays the output of a pip install command, showing that various packages were installed from cached wheels. The packages listed include typing_extensions, six, oauthlib, msgpack, idna, h11, certifi, repath, httpcore, anyio, httpx, and flet. The installation was successful, and the prompt is now '(venv) baycodec@Servidor: ~/Documentos/mi_app_flet\$'.

```
Using cached h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
Collecting six>=1.9.0 (from repath>=0.9.0->flet)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting typing_extensions>=4.5 (from anyio->httpx>=0.28.1->flet)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached flet-0.85.1-py3-none-any.whl (579 kB)
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (427 kB)
Using cached oauthlib-3.3.1-py3-none-any.whl (160 kB)
Using cached repath-0.9.0-py3-none-any.whl (4.7 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached idna-3.15-py3-none-any.whl (72 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna, h11, certifi, repath, httpcore, anyio, httpx, flet
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six-1.17.0 typing_extensions-4.15.0
(venv) baycodec@Servidor: ~/Documentos/mi_app_flet$
```

1. Crear el archivo de código

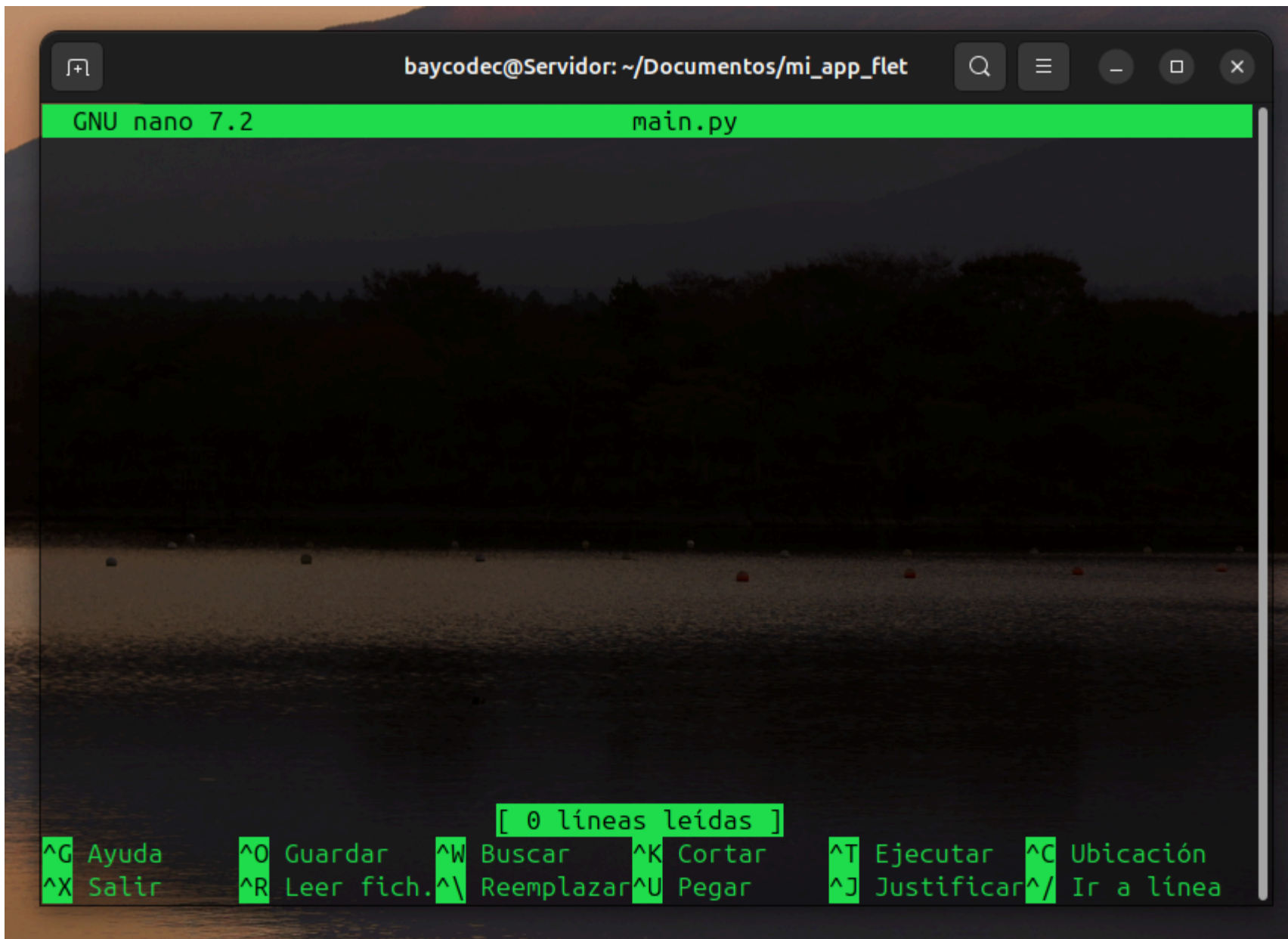
Para crear un archivo vacío llamado `main.py`, ejecuta este comando:


```
baycodec@Servidor: ~/Documentos/mi_app_flet
Collecting six>=1.9.0 (from repath>=0.9.0->flet)
  Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting typing_extensions>=4.5 (from anyio->httpx>=0.28.1->flet)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached flet-0.85.1-py3-none-any.whl (579 kB)
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (427 kB)
Using cached oauthlib-3.3.1-py3-none-any.whl (160 kB)
Using cached repath-0.9.0-py3-none-any.whl (4.7 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached idna-3.15-py3-none-any.whl (72 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna, h11, certifi, repath, httpcore, anyio, httpx, flet
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six-1.17.0 typing_extensions-4.15.0
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ touch main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$
```

```
baycodec@Servidor: ~/Documentos/mi_app_flet
Collecting typing_extensions>=4.5 (from anyio->httpx>=0.28.1->flet)
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached flet-0.85.1-py3-none-any.whl (579 kB)
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (427 kB)
Using cached oauthlib-3.3.1-py3-none-any.whl (160 kB)
Using cached repath-0.9.0-py3-none-any.whl (4.7 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached idna-3.15-py3-none-any.whl (72 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna, h11, certifi, repath, httpcore, anyio, httpx, flet
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six-1.17.0 typing_extensions-4.15.0
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ touch main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ ls
main.py  venv
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$
```



```
baycodec@Servidor: ~/Documentos/mi_app_flet
Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached flet-0.85.1-py3-none-any.whl (579 kB)
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (427 kB)
Using cached oauthlib-3.3.1-py3-none-any.whl (160 kB)
Using cached repath-0.9.0-py3-none-any.whl (4.7 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached idna-3.15-py3-none-any.whl (72 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna, h11, certifi, repath, httpcore, anyio, httpx, flet
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six-1.17.0 typing_extensions-4.15.0
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ touch main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ ls
main.py  venv
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ nano main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$
```





```
baycodec@Servidor: ~/Documentos/mi_app_flet
GNU nano 7.2 main.py *
import flet as ft

def main(page: ft.Page):
    # Configurar el título de la ventana
    page.title = "Mi Primer Hola Mundo"

    # Crear un texto flotante
    saludo = ft.Text(";Hola Mundo desde Flet!", size=30, color="blue")

    # Agregar el texto a la pantalla
    page.add(saludo)

# Ejecutar como aplicación de escritorio
ft.app(target=main)

^G Ayuda      ^O Guardar    ^W Buscar     ^K Cortar     ^T Ejecutar   ^C Ubicación
^X Salir      ^R Leer fich. ^\ Reemplazar ^U Pegar      ^J Justificar ^/ Ir a línea
```

- Para guardar en nano: Presiona `Ctrl + O`, luego dale a `Enter`.
- Para salir de nano: Presiona `Ctrl + X`.

baycodec@Servidor: ~/Documentos/mi_app_flet

GNU nano 7.2 main.py *

```
import flet as ft

def main(page: ft.Page):
    # Configurar el título de la ventana
    page.title = "Mi Primer Hola Mundo"

    # Crear un texto flotante
    saludo = ft.Text("¡Hola Mundo desde Flet!", size=30, color="blue")

    # Agregar el texto a la pantalla
    page.add(saludo)

# Ejecutar como aplicación de escritorio
ft.app(target=main)
```

Nombre del archivo a escribir: main.py

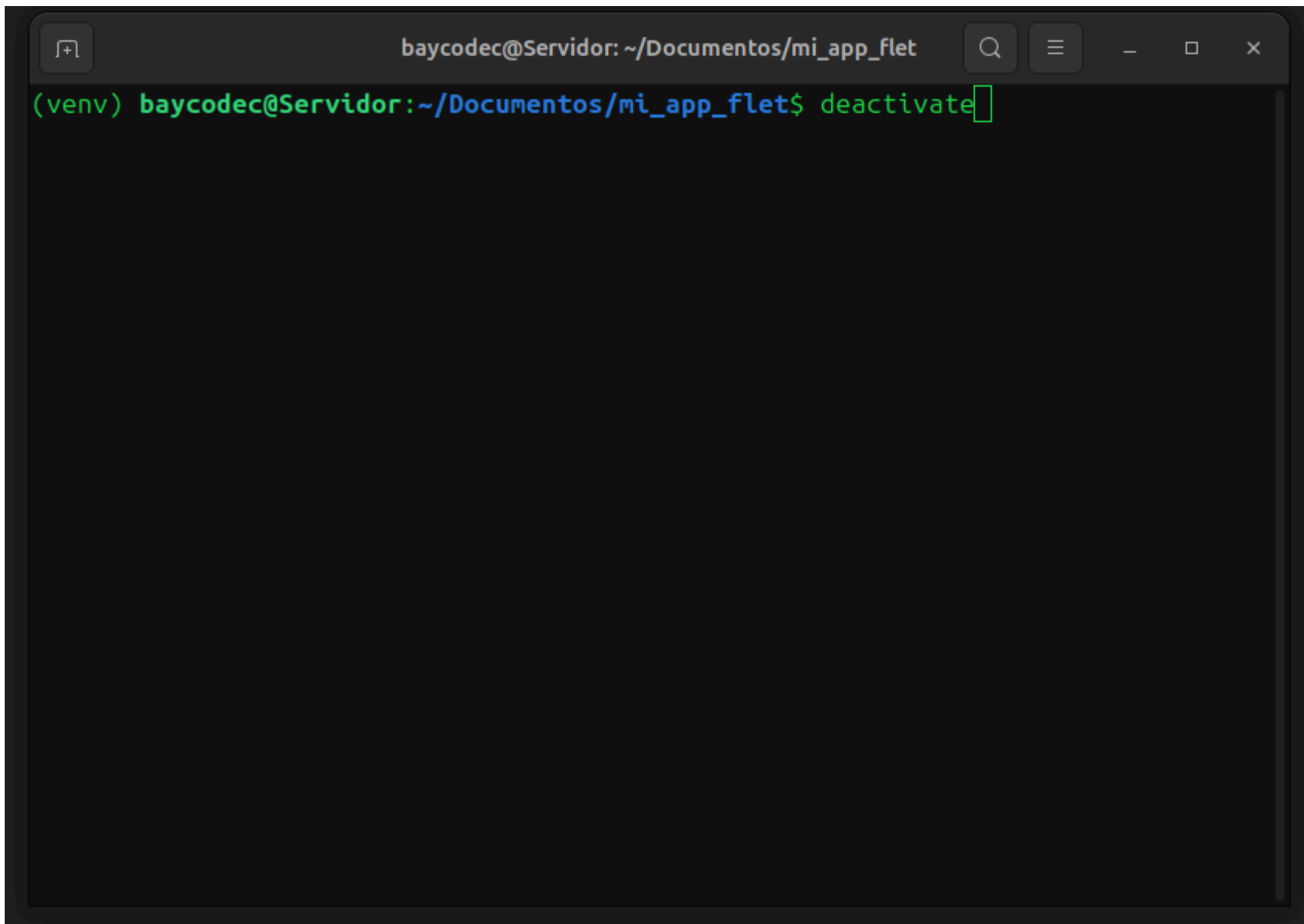
^G Ayuda	M-D Formato DOS	M-A Añadir	M-B Respalдар ficher
^C Cancelar	M-M Formato Mac	M-P Anteponer	^T Navegar

```
baycodec@Servidor: ~/Documentos/mi_app_flet
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (427 kB)
Using cached oauthlib-3.3.1-py3-none-any.whl (160 kB)
Using cached repath-0.9.0-py3-none-any.whl (4.7 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached idna-3.15-py3-none-any.whl (72 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna, h11, certifi, repath, httpcore, anyio, httpx, flet
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six-1.17.0 typing_extensions-4.15.0
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ touch main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ ls
main.py  venv
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ nano main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ nano main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ nano main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ python3 main.py
```

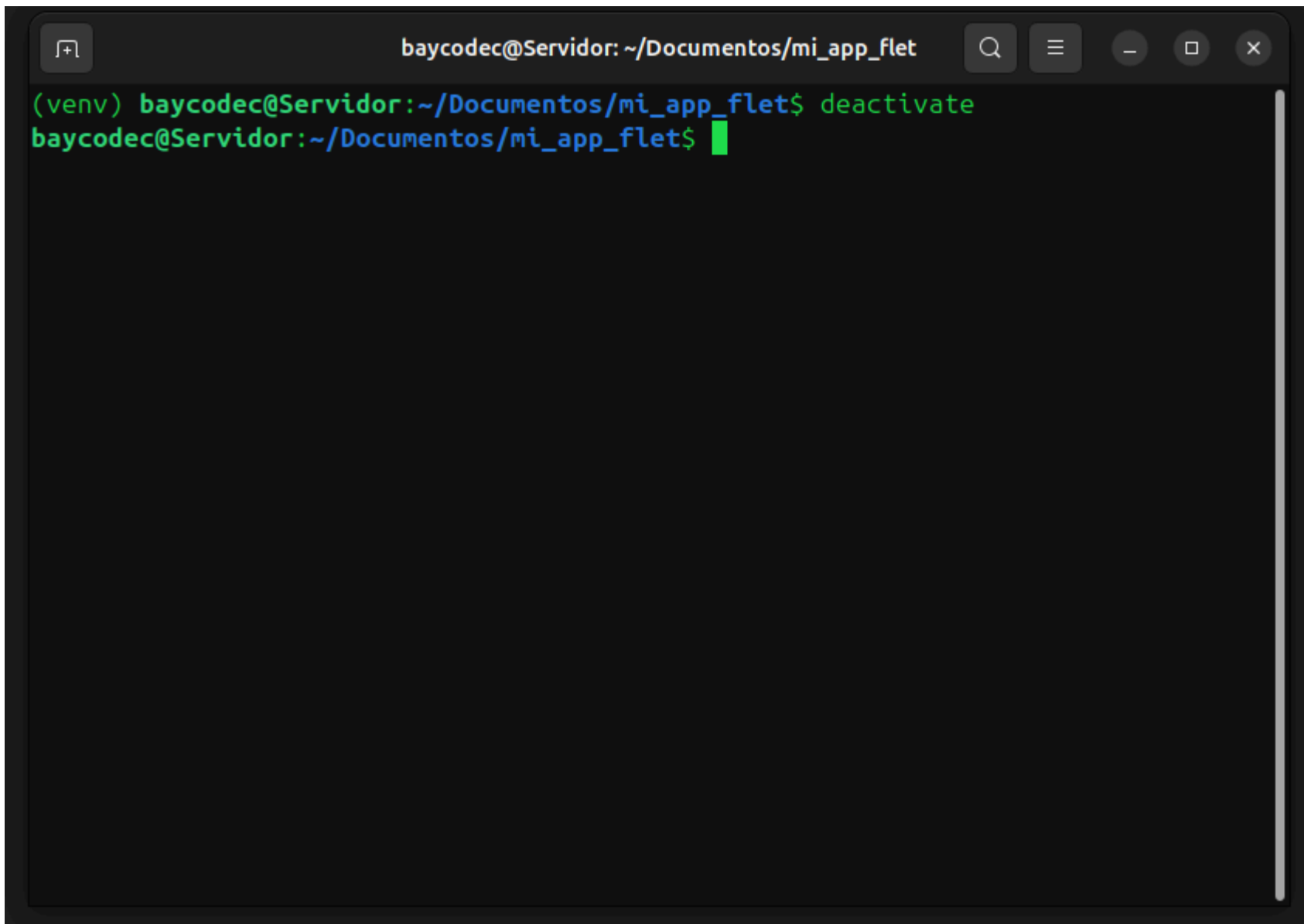

Mi Primer Hola Mundo

¡Hola Mundo desde Flet!

```
baycodec@Servidor: ~/Documentos/mi_app_flet
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached idna-3.15-py3-none-any.whl (72 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna,
h11, certifi, repath, httpcore, anyio, httpx, flet
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 htt
pcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six
-1.17.0 typing_extensions-4.15.0
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ touch main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ ls
main.py  venv
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ nano main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ nano main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ nano main.py
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ python3 main.py
/home/baycodec/Documentos/mi_app_flet/main.py:14: DeprecationWarning: app() is d
eprecated since version 0.80.0. Use run() instead.
  ft.app(target=main)
Installing flet-desktop 0.85.1 package...OK
Preparing Flet v0.85.1 for the first use. This is a one-time operation...
Gdk-Message: 09:27:25.621: Unable to load  from the cursor theme
█
```

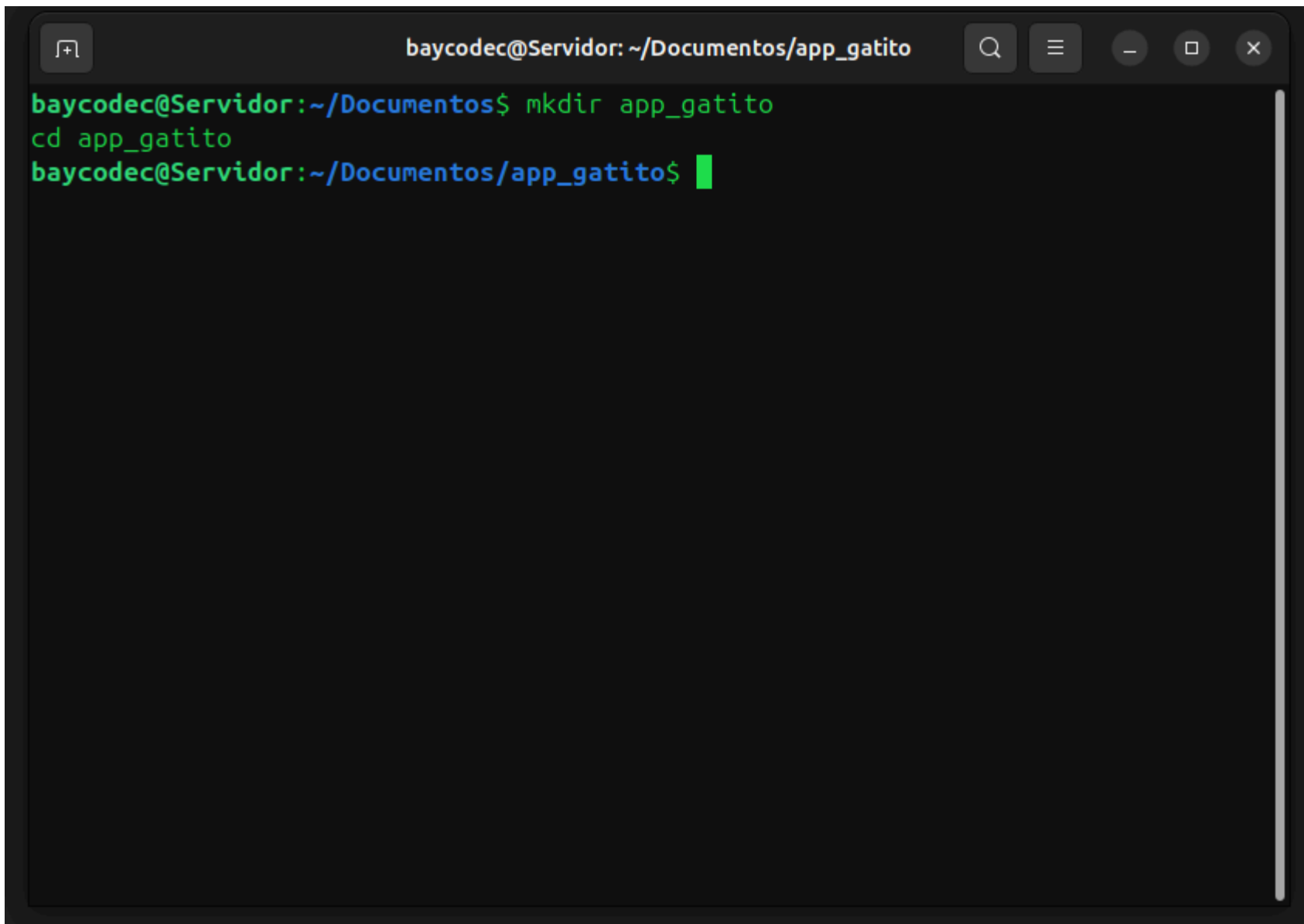

A terminal window with a dark background. The title bar at the top shows the user 'baycodec' on a server named 'Servidor', with the current directory being '~/Documentos/mi_app_flet'. There are standard window control buttons (search, menu, zoom, close) on the right. The terminal content shows a green prompt '(venv) baycodec@Servidor:~/Documentos/mi_app_flet\$' followed by the command 'deactivate' in green, with a green cursor at the end.

```
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ deactivate
```

A terminal window with a dark background. The title bar at the top shows the user 'baycodec' on a server named 'Servidor', with the current directory being '~/Documentos/mi_app_flet'. There are standard window control buttons (search, menu, zoom, maximize, close) on the right. The terminal content shows a green prompt '(venv) baycodec@Servidor:~/Documentos/mi_app_flet\$' followed by the command 'deactivate' in green. The next line shows the prompt 'baycodec@Servidor:~/Documentos/mi_app_flet\$' with a green cursor, indicating the command has been executed.

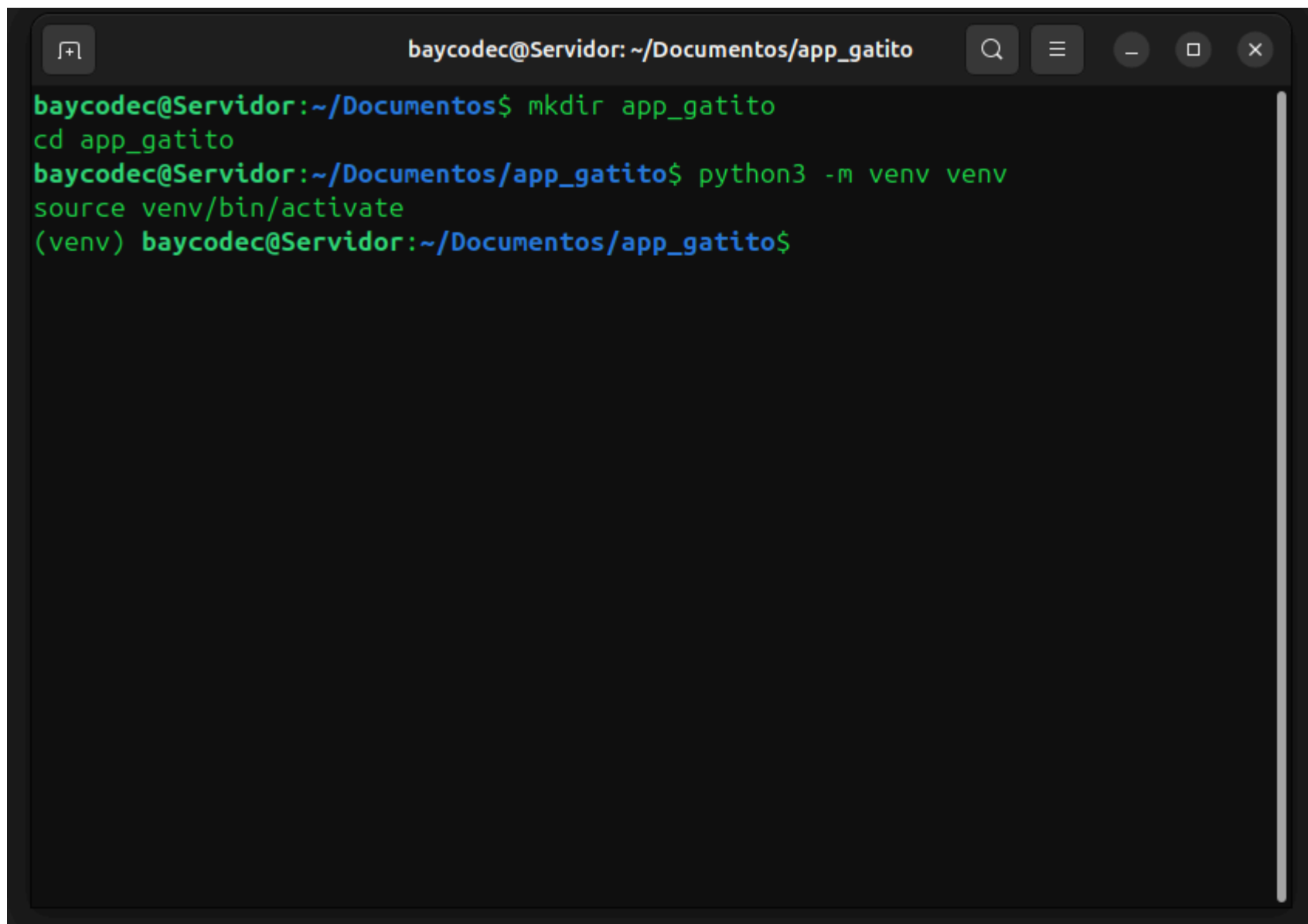
```
baycodec@Servidor: ~/Documentos/mi_app_flet
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ deactivate
baycodec@Servidor:~/Documentos/mi_app_flet$
```

```
baycodec@Servidor: ~/Documentos
(venv) baycodec@Servidor:~/Documentos/mi_app_flet$ deactivate
baycodec@Servidor:~/Documentos/mi_app_flet$ cd --.
bash: cd: --: opción no válida
cd: modo de empleo: cd [-L|[-P [-e]]] [dir]
baycodec@Servidor:~/Documentos/mi_app_flet$ cd ..
baycodec@Servidor:~/Documentos$
```



A terminal window with a dark background. The title bar at the top shows the user 'baycodec' on a server named 'Servidor', with the current directory being '~/.Documentos/app_gatito'. The window includes standard macOS window controls (zoom, search, menu, and window management buttons). The terminal content shows the user creating a directory 'app_gatito' and then navigating into it. The prompt changes from '~/.Documentos\$' to '~/.Documentos/app_gatito\$' after the 'cd' command.

```
baycodec@Servidor: ~/.Documentos/app_gatito
baycodec@Servidor:~/.Documentos$ mkdir app_gatito
cd app_gatito
baycodec@Servidor:~/.Documentos/app_gatito$
```

A terminal window with a dark background and light green text. The window title bar shows the user 'baycodec' on a server named 'Servidor', with the current directory being '~/.Documentos/app_gatito'. The terminal contains the following commands and output:

```
baycodec@Servidor:~/Documentos$ mkdir app_gatito
cd app_gatito
baycodec@Servidor:~/Documentos/app_gatito$ python3 -m venv venv
source venv/bin/activate
(venv) baycodec@Servidor:~/Documentos/app_gatito$
```

The window includes standard macOS-style window controls (zoom, search, menu, and window management buttons) in the top right corner.

```
baycodec@Servidor: ~/Documentos/app_gatito
cd app_gatito
baycodec@Servidor:~/Documentos/app_gatito$ python3 -m venv venv
source venv/bin/activate
(venv) baycodec@Servidor:~/Documentos/app_gatito$ pip install flet
Collecting flet
  Using cached flet-0.85.1-py3-none-any.whl.metadata (6.9 kB)
Collecting oauthlib>=3.2.2 (from flet)
  Using cached oauthlib-3.3.1-py3-none-any.whl.metadata (7.9 kB)
Collecting httpx>=0.28.1 (from flet)
  Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting repath>=0.9.0 (from flet)
  Using cached repath-0.9.0-py3-none-any.whl.metadata (899 bytes)
Collecting msgpack>=1.1.0 (from flet)
  Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (8.1 kB)
Collecting anyio (from httpx>=0.28.1->flet)
  Using cached anyio-4.13.0-py3-none-any.whl.metadata (4.5 kB)
Collecting certifi (from httpx>=0.28.1->flet)
  Using cached certifi-2026.5.20-py3-none-any.whl.metadata (2.5 kB)
Collecting httpcore==1.* (from httpx>=0.28.1->flet)
  Using cached httpcore-1.0.9-py3-none-any.whl.metadata (21 kB)
Collecting idna (from httpx>=0.28.1->flet)
  Using cached idna-3.15-py3-none-any.whl.metadata (7.7 kB)
Collecting h11>=0.16 (from httpcore==1.*->httpx>=0.28.1->flet)
```

```
baycodec@Servidor: ~/Documentos/app_gatito
Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting typing_extensions>=4.5 (from anyio->httpx>=0.28.1->flet)
Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached flet-0.85.1-py3-none-any.whl (579 kB)
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (427 kB)
Using cached oauthlib-3.3.1-py3-none-any.whl (160 kB)
Using cached repath-0.9.0-py3-none-any.whl (4.7 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached idna-3.15-py3-none-any.whl (72 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna, h11, certifi, repath, httpcore, anyio, httpx, flet
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six-1.17.0 typing_extensions-4.15.0
(venv) baycodec@Servidor:~/Documentos/app_gatito$ ls
venv
(venv) baycodec@Servidor:~/Documentos/app_gatito$
```

```
baycodec@Servidor: ~/Documentos/app_gatito
Using cached six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting typing_extensions>=4.5 (from anyio->httpx>=0.28.1->flet)
Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Using cached flet-0.85.1-py3-none-any.whl (579 kB)
Using cached httpx-0.28.1-py3-none-any.whl (73 kB)
Using cached httpcore-1.0.9-py3-none-any.whl (78 kB)
Using cached msgpack-1.1.2-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (427 kB)
Using cached oauthlib-3.3.1-py3-none-any.whl (160 kB)
Using cached repath-0.9.0-py3-none-any.whl (4.7 kB)
Using cached six-1.17.0-py2.py3-none-any.whl (11 kB)
Using cached anyio-4.13.0-py3-none-any.whl (114 kB)
Using cached idna-3.15-py3-none-any.whl (72 kB)
Using cached certifi-2026.5.20-py3-none-any.whl (134 kB)
Using cached h11-0.16.0-py3-none-any.whl (37 kB)
Using cached typing_extensions-4.15.0-py3-none-any.whl (44 kB)
Installing collected packages: typing_extensions, six, oauthlib, msgpack, idna, h11, certifi, repath, httpcore, anyio, httpx, flet
Successfully installed anyio-4.13.0 certifi-2026.5.20 flet-0.85.1 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.15 msgpack-1.1.2 oauthlib-3.3.1 repath-0.9.0 six-1.17.0 typing_extensions-4.15.0
(venv) baycodec@Servidor:~/Documentos/app_gatito$ ls
venv
(venv) baycodec@Servidor:~/Documentos/app_gatito$ mkdir assets
```



```
baycodec@Servidor: ~/Documentos/app_gatito
(venv) baycodec@Servidor:~/Documentos/app_gatito$ ls
assets  venv
(venv) baycodec@Servidor:~/Documentos/app_gatito$ tree assets/
assets/
├── gatito.mp3
└── gatito.png

1 directory, 2 files
(venv) baycodec@Servidor:~/Documentos/app_gatito$
```



gatito.mp3



gatito.png



baycodec@Servidor: ~/Documentos/app_gatito



```
(venv) baycodec@Servidor:~/Documentos/app_gatito$ ls
assets  venv
(venv) baycodec@Servidor:~/Documentos/app_gatito$ tree assets/
assets/
├── gatito.mp3
└── gatito.png

1 directory, 2 files
(venv) baycodec@Servidor:~/Documentos/app_gatito$
```



gatito.mp3



gatito.png



baycodec@Servidor: ~/Documentos/app_gatito

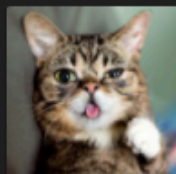


```
(venv) baycodec@Servidor:~/Documentos/app_gatito$ ls
assets  venv
(venv) baycodec@Servidor:~/Documentos/app_gatito$ tree assets/
assets/
├── gatito.mp3
└── gatito.png

1 directory, 2 files
(venv) baycodec@Servidor:~/Documentos/app_gatito$ ls
assets  venv
(venv) baycodec@Servidor:~/Documentos/app_gatito$ nano main.py
```



gatito.mp3



gatito.png

```
GNU nano 7.2 main.py *
import flet as ft

def main(page: ft.Page):
    page.title = "El Gatito Que Maúlla"
    page.vertical_alignment = ft.MainAxisAlignment.CENTER
    page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
    page.bgcolor = ft.colors.BLUE_GREY_900 # Un fondo oscuro para que resalte

    # 1. Configurar el reproductor de audio apuntando a tu archivo mp3
    audio_gatito = ft.Audio(
        src="gatito.mp3", # Flet busca automáticamente dentro de la carpeta assets
        autoplay=False
    )
    # Es obligatorio agregar el control de audio a la página (aunque no sea visible)
    page.overlay.append(audio_gatito)

    # 2. Definir la función que se ejecutará al hacer clic
    def maullar(e):
        audio_gatito.play() # Reproduce el sonido

    # 3. Crear el botón contenedor que tiene la imagen dentro
    boton_gatito = ft.GestureDetector(
        content=ft.Image(
            src="gatito.png", # Tu imagen en assets
            width=300,
            height=300,
            fit=ft.ImageFit.CONTAIN,
            border_radius=ft.border_radius.all(20) # Bordes redondeados estéticos
        ),
        on_tap=maullar, # Al tocar la imagen, llama a la función
        mouse_cursor=ft.MouseCursor.CLICK # Cambia el cursor a una manito en PC
    )

    # 4. Agregar un texto de instrucción y el botón a la pantalla
    page.add(
        ft.Text(";Presiona al gatito para escucharlo!", size=20, color="white", weight=ft.FontWeight.BOLD),
        ft.Container(height=20), # Espacio en blanco
        boton_gatito
    )
```

^G Ayuda

^O Guardar

^W Buscar

^K Cortar

^T Ejecutar

^C Ubicación

M-U Deshacer

M-A Poner marca

^X Salir

^R Leer fich.

^ Reemplazar

^U Pegar

^J Justificar

^/ Ir a línea

M-E Rehacer

M-6 Copiar

FOLDERS

▼ app_gatito
 ▶ assets
 ▶ venv
 main.py.swp
/* main.py

```
main.py x
1 import flet as ft
2
3 def main(page: ft.Page):
4     page.title = "El Gatito Que Maúlla"
5     page.vertical_alignment = ft.MainAxisAlignment.CENTER
6     page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
7     page.bgcolor = ft.colors.BLUE_GREY_900 # Un fondo oscuro para que resalte
8
9     # 1. Configurar el reproductor de audio apuntando a tu archivo mp3
10    audio_gatito = ft.Audio(
11        src="gatito.mp3", # Flet busca automáticamente dentro de la carpeta assets
12        autoplay=False
13    )
14    # Es obligatorio agregar el control de audio a la página (aunque no sea visible)
15    page.overlay.append(audio_gatito)
16
17    # 2. Definir la función que se ejecutará al hacer clic
18    def maullar(e):
19        audio_gatito.play() # Reproduce el sonido
20
21    # 3. Crear el botón contenedor que tiene la imagen dentro
22    boton_gatito = ft.GestureDetector(
23        content=ft.Image(
24            src="gatito.png", # Tu imagen en assets
25            width=300,
26            height=300,
27            fit=ft.ImageFit.CONTAIN,
28            border_radius=ft.border_radius.all(20) # Bordes redondeados estéticos
29        ),
30        on_tap=maullar, # Al tocar la imagen, llama a la función
31        mouse_cursor=ft.MouseCursor.CLICK # Cambia el cursor a una manito en PC
32    )
33
34    # 4. Agregar un texto de instrucción y el botón a la pantalla
35    page.add(
36        ft.Text("¡Presiona al gatito para escucharlo!", size=20, color="white", weight=ft.FontWeight.BOLD),
37        ft.Container(height=20), # Espacio en blanco
38        boton_gatito
39    )
40
41    # Indicarle a Flet dónde está la carpeta de recursos (assets)
42    ft.app(target=main, assets_dir="assets")
43
```



```
~/Documentos/app_gatito/main.py (app_gatito) - Sublime Text (UNREGISTERED)
Archivo Editar Selección Buscar Mostrar Ira... Herramientas Proyecto Preferencias Ayuda

FOLDERS
└─ app_gatito
  └─ assets
  └─ venv
  /* main.py

main.py
1  import flet as ft
2
3  def main(page: ft.Page):
4      page.title = "El Gatito Que Maúlla"
5      page.vertical_alignment = ft.MainAxisAlignment.CENTER
6      page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
7      page.bgcolor = ft.colors.BLUE_GREY_900 # Un fondo oscuro para que resalte
8
9      # 1. Configurar el reproductor de audio apuntando a tu archivo mp3
10     audio_gatito = ft.Audio(
11         src="gatito.mp3", # Flet busca automáticamente dentro de la carpeta assets
12         autoplay=False
13     )
14     # Es obligatorio agregar el control de audio a la página (aunque no sea visible)
15     page.overlay.append(audio_gatito)
16
17     # 2. Definir la función que se ejecutará al hacer clic
18     def maullar(e):
19         audio_gatito.play() # Reproduce el sonido
20
21     # 3. Crear el botón contenedor que tiene la imagen dentro
22     boton_gatito = ft.GestureDetector(
23         content=ft.Image(
24             src="gatito.png", # Tu imagen en assets
25             width=300,
26             height=300,
27             fit=ft.ImageFit.CONTAIN,
28             border_radius=ft.border_radius.all(20) # Bordes redondeados estéticos
29         ),
30         on_tap=maullar, # Al tocar la imagen, llama a la función
31         mouse_cursor=ft.MouseCursor.CLICK # Cambia el cursor a una manito en PC
32     )
33
34     # 4. Agregar un texto de instrucción y el botón a la pantalla
35     page.add(
36         ft.Text("¡Presiona al gatito para escucharlo!", size=20, color="white", weight=ft.FontWeight.BOLD),
37         ft.Container(height=20), # Espacio en blanco
38         boton_gatito
39     )
40
41     # Indicarle a Flet dónde está la carpeta de recursos (assets)
42     ft.app(target=main, assets_dir="assets")
43
```

1 character selected Spaces: 4 Python

page.bgcolor = ft.Colors.BLUE_GREY_900

FOLDERS

▼ app_gatito
 ▶ assets
 ▶ venv
/* main.py

```
main.py
1  import flet as ft
2
3  def main(page: ft.Page):
4      page.title = "El Gatito Que Maúlla"
5      page.vertical_alignment = ft.MainAxisAlignment.CENTER
6      page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
7      page.bgcolor = ft.colors.BLUE_GREY_900 # Un fondo oscuro para que resalte
8
9      # 1. Configurar el reproductor de audio apuntando a tu archivo mp3
10     audio_gatito = ft.Audio(
11         src="gatito.mp3", # Flet busca automáticamente dentro de la carpeta assets
12         autoplay=False
13     )
14     # Es obligatorio agregar el control de audio a la página (aunque no sea visible)
15     page.overlay.append(audio_gatito)
16
17     # 2. Definir la función que se ejecutará al hacer clic
18     def maullar(e):
19         audio_gatito.play() # Reproduce el sonido
20
21     # 3. Crear el botón contenedor que tiene la imagen dentro
22     boton_gatito = ft.GestureDetector(
23         content=ft.Image(
24             src="gatito.png", # Tu imagen en assets
25             width=300,
26             height=300,
27             fit=ft.ImageFit.CONTAIN,
28             border_radius=ft.border_radius.all(20) # Bordes redondeados estéticos
29         ),
30         on_tap=maullar, # Al tocar la imagen, llama a la función
31         mouse_cursor=ft.MouseCursor.CLICK # Cambia el cursor a una manito en PC
32     )
33
34     # 4. Agregar un texto de instrucción y el botón a la pantalla
35     page.add(
36         ft.Text("¡Presiona al gatito para escucharlo!", size=20, color="white", weight=ft.FontWeight.BOLD),
37         ft.Container(height=20), # Espacio en blanco
38         boton_gatito
39     )
40
41     # Indicarle a Flet dónde está la carpeta de recursos (assets)
42     ft.app(target=main, assets_dir="assets")
43
```

Archivo Editar Selección Buscar Mostrar Ira... Herramientas Proyecto Preferencias Ayuda

FOLDERS

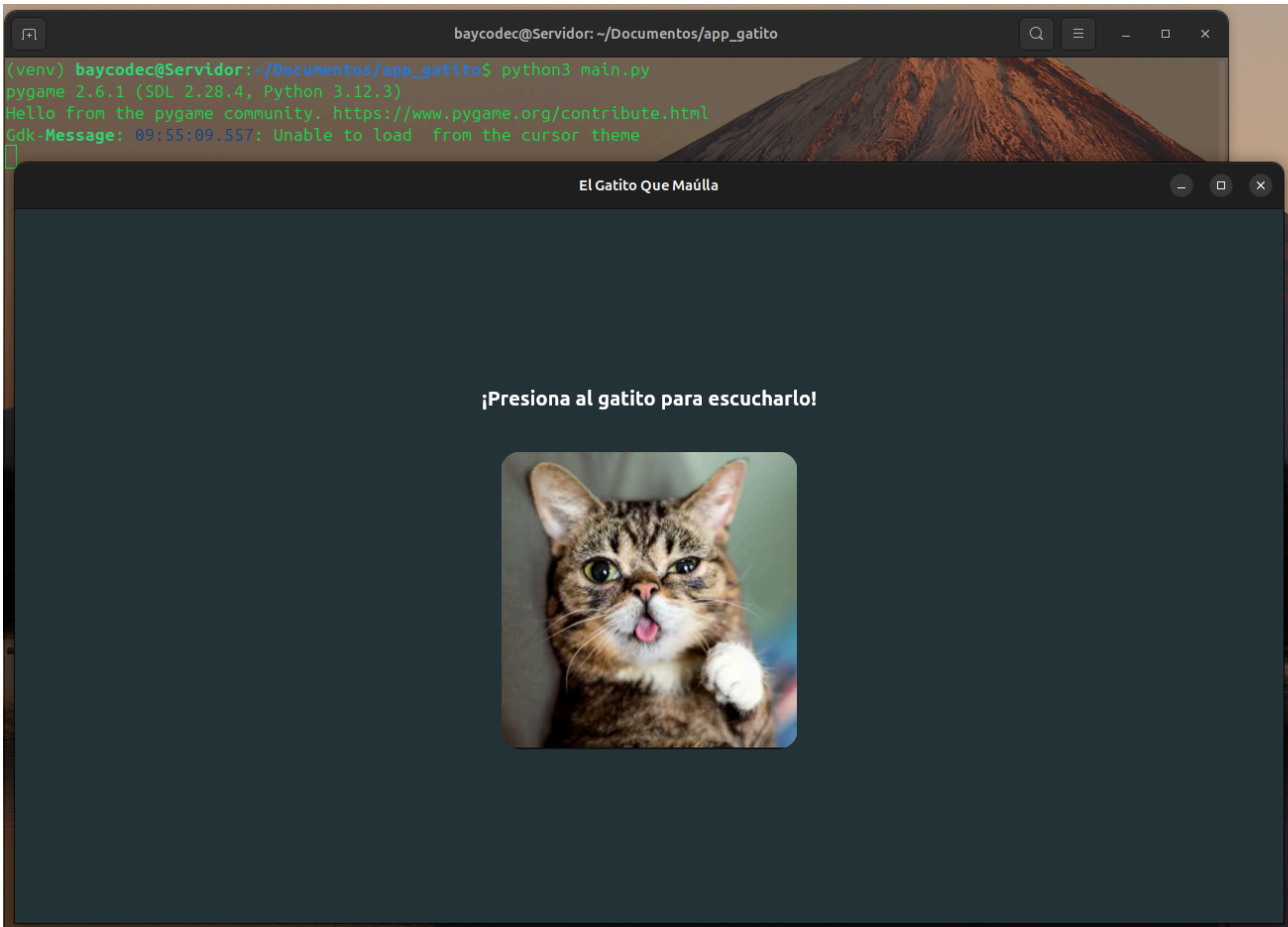
- app_gatito
 - assets
 - venv
- /* main.py

```
main.py x
1  import flet as ft
2
3  def main(page: ft.Page):
4      page.title = "El Gatito Que Maúlla"
5      page.vertical_alignment = ft.MainAxisAlignment.CENTER
6      page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
7      page.bgcolor = ft.colors.BLUE_GREY_900 # Un fondo oscuro para que resalte
8
9      # 1. Configurar el reproductor de audio apuntando a tu archivo mp3
10     audio_gatito = ft.Audio(
11         src="gatito.mp3", # Flet busca automáticamente dentro de la carpeta assets
12         autoplay=False
13     )
14     # Es obligatorio agregar el control de audio a la página (aunque no sea visible)
15     page.overlay.append(audio_gatito)
16
17     # 2. Definir la función que se ejecutará al hacer clic
18     def maullar(e):
19         audio_gatito.play() # Reproduce el sonido
20
21     # 3. Crear el botón contenedor que tiene la imagen dentro
22     boton_gatito = ft.GestureDetector(
23         content=ft.Image(
24             src="gatito.png", # Tu imagen en assets
25             width=300,
26             height=300,
27             fit=ft.ImageFit.CONTAIN,
28             border_radius=ft.border_radius.all(20) # Bordes redondeados estéticos
29         ),
30         on_tap=maullar, # Al tocar la imagen, llama a la función
31         mouse_cursor=ft.MouseCursor.CLICK # Cambia el cursor a una manito en PC
32     )
33
34     # 4. Agregar un texto de instrucción y el botón a la pantalla
35     page.add(
36         ft.Text("¡Presiona al gatito para escucharlo!", size=20, color="white", weight=ft.FontWeight.BOLD),
37         ft.Container(height=20), # Espacio en blanco
38         boton_gatito
39     )
40
41     # Indicarle a Flet dónde está la carpeta de recursos (assets)
42     ft.run(target=main, assets_dir="assets")
43
```



```
baycodec@Servidor: ~/Documentos/app_gatito
(venv) baycodec@Servidor:~/Documentos/app_gatito$ pip install --upgrade flet
Requirement already satisfied: flet in ./venv/lib/python3.12/site-packages (0.85.1)
Requirement already satisfied: oauthlib>=3.2.2 in ./venv/lib/python3.12/site-packages (from flet) (3.3.1)
Requirement already satisfied: httpx>=0.28.1 in ./venv/lib/python3.12/site-packages (from flet) (0.28.1)
Requirement already satisfied: repath>=0.9.0 in ./venv/lib/python3.12/site-packages (from flet) (0.9.0)
Requirement already satisfied: msgpack>=1.1.0 in ./venv/lib/python3.12/site-packages (from flet) (1.1.2)
Requirement already satisfied: anyio in ./venv/lib/python3.12/site-packages (from httpx>=0.28.1->flet) (4.13.0)
Requirement already satisfied: certifi in ./venv/lib/python3.12/site-packages (from httpx>=0.28.1->flet) (2026.5.20)
Requirement already satisfied: httpcore==1.* in ./venv/lib/python3.12/site-packages (from httpx>=0.28.1->flet) (1.0.9)
Requirement already satisfied: idna in ./venv/lib/python3.12/site-packages (from httpx>=0.28.1->flet) (3.15)
Requirement already satisfied: h11>=0.16 in ./venv/lib/python3.12/site-packages (from httpcore==1.*->httpx>=0.28.1->flet) (0.16.0)
Requirement already satisfied: six>=1.9.0 in ./venv/lib/python3.12/site-packages (from repath>=0.9.0->flet) (1.17.0)
Requirement already satisfied: typing_extensions>=4.5 in ./venv/lib/python3.12/site-packages (from anyio->httpx>=0.28.1->flet) (4.15.0)
(venv) baycodec@Servidor:~/Documentos/app_gatito$
```

```
baycodec@Servidor: ~/Documentos/app_gatito
(venv) baycodec@Servidor:~/Documentos/app_gatito$ pip install pygame
Collecting pygame
  Downloading pygame-2.6.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (12 kB)
  Downloading pygame-2.6.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (14.0 MB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 14.0/14.0 MB 56.3 MB/s eta 0:00:00
Installing collected packages: pygame
Successfully installed pygame-2.6.1
(venv) baycodec@Servidor:~/Documentos/app_gatito$
```



FOLDERS

▼ app_gatito
 ▶ assets
 ▶ venv
/* main.py

```
main.py
1  import flet as ft
2  import os
3  from pygame import mixer # Usamos el motor de audio nativo de Python
4
5  def main(page: ft.Page):
6      page.title = "El Gatito Que Maúlla"
7      page.vertical_alignment = ft.MainAxisAlignment.CENTER
8      page.horizontal_alignment = ft.CrossAxisAlignment.CENTER
9      page.bgcolor = ft.Colors.BLUE_GREY_900
10
11     # 1. Inicializar el motor de audio de pygame
12     mixer.init()
13
14     # Construir la ruta exacta al archivo mp3 dentro de assets
15     ruta_audio = os.path.join("assets", "gatito.mp3")
16
17     # 2. Definir la función que reproducirá el sonido al hacer clic
18     def maullar(e):
19         try:
20             mixer.music.load(ruta_audio)
21             mixer.music.play()
22         except Exception as ex:
23             print(f"Error al reproducir audio: {ex}")
24
25     # 3. Crear el botón contenedor con la imagen del gatito
26     boton_gatito = ft.GestureDetector(
27         content=ft.Image(
28             src="gatito.png",
29             width=300,
30             height=300,
31             fit="contain", # Compatible con todas las versiones
32             border_radius=ft.BorderRadius.all(20)
33         ),
34         on_tap=maullar,
35         mouse_cursor=ft.MouseCursor.CLICK
36     )
37
38     # 4. Agregar componentes a la pantalla
39     page.add(
40         ft.Text("¡Presiona al gatito para escucharlo!", size=20, color="white", weight=ft.FontWeight.BOLD),
41         ft.Container(height=20,
42             boton_gatito
43         )
44
45     # Ejecutar la aplicación indicando la carpeta de assets
46     ft.run(main, assets_dir="assets")
```

El Gatito Que Maúlla

¡Presiona al gatito para escucharlo!

